



Retina Blood Vessels Segmentation and Classification with the Multi-featured Approach

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Abstract

Segmenting retinal blood vessels poses a significant challenge due to the irregularities inherent in small vessels. The complexity arises from the intricate task of effectively merging features at multiple levels, coupled with potential spatial information loss during successive down-sampling steps. This particularly affects the identification of small and faintly contrasting vessels. To address these challenges, we present a model tailored for automated arterial and venous (A/V) classification, complementing blood vessel segmentation. This paper presents an advanced methodology for segmenting and classifying retinal vessels using a series of sophisticated pre-processing and feature extraction techniques. The ensemble filter approach, incorporating Bilateral and Laplacian edge detectors, enhances image contrast and preserves edges. The proposed algorithm further refines the image by generating an orientation map. During the vessel extraction step, a complete convolution network processes the input image to create a detailed vessel map, enhanced by attention operations that improve modeling perception and resilience. The encoder extracts semantic features, while the Attention Module refines blood vessel depiction, resulting in highly accurate segmentation outcomes. The model was verified using the STARE dataset, which includes 400 images; the DRIVE dataset with 40 images; the HRF dataset with 45 images; and the INSPIRE-AVR dataset containing 40 images. The proposed model demonstrated superior performance across all datasets, achieving an accuracy of 97.5% on the DRIVE dataset, 99.25% on the STARE dataset, 98.33% on the INSPIRE-AVR dataset, and 98.67% on the HRF dataset. These results highlight the method's effectiveness in accurately segmenting and classifying retinal vessels.

Keywords A/V classification · Multi-level features · Parallel channel attention · Tiny vessels

Introduction

Common eye diseases, including hypertensive retinopathy (HR), diabetic retinopathy (DR), and glaucoma, predominantly originate from alterations in the blood vessels of the retina—a light-sensitive membrane [1]. Prolonged hypertension can result in arteriovenous nicking and systemic arteriolar stenosis [2]. Diabetes, on the other hand, may lead to the formation of numerous abnormal blood vessels prone to rupture [3]. Additionally, the avascular macular region, typically devoid of blood vessels, can experience the undesirable development of new blood vessels [4].

The implementation of computer-aided diagnosis (CAD) systems in ophthalmology is crucial for enhancing diagnostic accuracy and efficiency, particularly in identifying eye conditions related to vascular alterations and arteriovenous (A/V) states [5]. These systems leverage advanced image processing algorithms and machine learning techniques to automatically detect and delineate retinal blood vessels from fundus images. CAD systems provide quantitative assessments of vessel characteristics such as diameter, tortuosity, and crossing patterns, essential for early detection of diseases like diabetic retinopathy and hypertensive retinopathy.

In clinical settings, there is a strong preference for automated segmentation and classification due to the labor-intensive and subjective nature of manual identification in fundus images [6–8]. Expert segmentation can vary significantly [9–12], underscoring the need for automated, highly precise techniques in retinal vascular segmentation. The complex morphological structure and abundance of tiny vessels, often with widths of one pixel or fewer than ten pixels, present

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challenges in accurately separating the retinal vascular tree from the background. CAD systems alleviate variability and subjectivity, ensuring consistent and reliable results across different clinical settings while enabling timely intervention and treatment planning by monitoring disease progression with greater precision and confidence [13].

Likewise, the presence of uneven illumination and lesion zones introduces challenges in differentiating non-vascular tissues from blood vessels, leading to minimal contrast. The precise segmentation of retinal vessels in fundus images remains a formidable task due to these factors, especially when dealing with small and low-contrast vessels. Microvessels, characterized by a limited number of pixels, play a crucial role in diagnosing neovascular disorders, emphasizing the importance of focusing on small vessels rather than larger ones. The process of identifying and separating distinct retinal structures, referred to as retinal segmentation, proves valuable in the diagnosis and treatment of various eye disorders [14].

Yang et al. [15] proposed a strategy that integrates U-Net with deformable convolution for identifying retinal blood vessels. Their model achieved sensitivity (SE), specificity (SP), accuracy (Acc), and area under the curve (AUC) of 0.8115, 0.9780, 0.9568, and 0.9810 on the DRIVE dataset, and 0.8075, 0.9841, 0.9664, and 0.9872 on CHASE_DB1, respectively. Guo et al. [16] introduced a generative adversarial network with U-Net for end-to-end processing to segment retinal vessels, achieving 0.9772 AUROC, 0.9058 AUPRC, and 0.8215 dice on the DRIVE dataset. Despite these advancements, the challenge of segmenting thin vessels and maintaining vessel connectivity persists due to inefficient use of contextual information. Vascular connections and thin blood vessels are crucial for diagnosing vascular disorders [17, 18]. By combining the ROI feature vector composition method with a neural network, they correctly classified 95.32% of vessel center lines between two crossings. The Swin-Unet model, featuring encoder and decoder with pure transformer modules and a symmetrical architecture similar to U-Net, has been proposed for medical image segmentation [19, 20], achieving accuracy, precision, recall, and F1 values of 98.65, 87.19, 88.63, and 87.20, respectively. Wen et al. [21] presented an approach for segmenting the optic disc and cup in the retina using a combination of CNN and transformer. Features are extracted through convolution and then processed through transformer and multi-scale convolution modules, enhancing segmentation performance. Extensive experiments showed improvements of about 7.6% in overall accuracy, 10.9% in sensitivity, and 9.2% in specificity compared to previous methods. Hong et al. [22] achieved multi-organ segmentation by integrating the transformer with the U-Net model, demonstrating a Dice similarity coefficient (DSC) of 80.68%. The

transformer module addresses global interactions after low-level feature extraction using convolutions, enabling more accurate and detailed organ segmentation. Several methods have attempted to address these challenges. Yang et al. proposed integrating U-Net with deformable convolution for retinal blood vessel identification, and Guo et al. introduced a generative adversarial network (GAN) with U-Net for end-to-end retinal vessel segmentation. While these methods have shown improvements in segmentation accuracy, they still struggle with thin vessel segmentation and vessel connection due to inefficient contextual information utilization. Accurately segmenting and maintaining the connectivity of thin blood vessels is vital for diagnosing vascular disorders, highlighting the need for innovative approaches in retinal vascular segmentation.

Despite recent advances and the introduction of numerous techniques aimed at mitigating existing drawbacks, arterial/venous (A/V) classification of retinal vessels remains a challenging issue. Firstly, the shape and architecture of vessels exhibit considerable variation owing to the intricate environment within the human eye. The curvature of blood vessels further intensifies the complexity of accurately segmenting and classifying them. Secondly, the subtle peripheral vascular differences make distinguishing between arteries and veins more challenging, as they exhibit minimal distinctions, particularly towards the extremities of blood vessels. These challenges are compounded by the difficulty in accurately segmenting thin vessels and maintaining vessel connectivity, both of which are crucial for clinical diagnostics.

In response to these challenges, we introduce a unique approach that minimizes feature map inconsistencies and enhances feature map compatibility between different layers through the utilization of a multi-scale features unit. Our proposed deep learning end-to-end technique for A/V classification and segmentation aims to alleviate the aforementioned issues. By leveraging an enhanced convolutional neural network (CNN) architecture combined with attention mechanisms, this approach improves upon the limitations of traditional methods by effectively capturing and utilizing contextual information to distinguish between arteries and veins, even with subtle differences. The encoding section converts input images into feature maps using convolutional layers, while retina vessel extraction focuses on identifying and extracting all blood vessels. A/V extraction goes a step further by classifying these vessels into arteries and veins through additional processing. The integration of convolutional networks with attention mechanisms further refines feature extraction, striking a balance between model complexity and interpretability. This design choice optimizes the model's efficiency in clinical settings, where both accuracy and practical usability are paramount. By incorporating multi-level attention mechanisms, the model enhances its ability to accurately segment thin vessels and maintain

vessel connectivity, thereby advancing the field of retinal vessel segmentation and classification.

Research Contributions

The primary contributions are enumerated as follows:

1. A novel approach to A/V classification is put forth, breaking the work into three distinct phases: vessel feature extraction, artery, and vein classification, and tuning.
2. An attention module is presented to selectively emphasize key aspects beneficial for a given activity while suppressing redundant, irrelevant information, therefore fusing multi-level features in an adaptive and effective manner.
3. We implemented a parallel channel and added it to the up sample to highlight the boundary data of small vessels.
4. Extensive experimental findings on a variety of public datasets demonstrate the novel approach's feasibility,

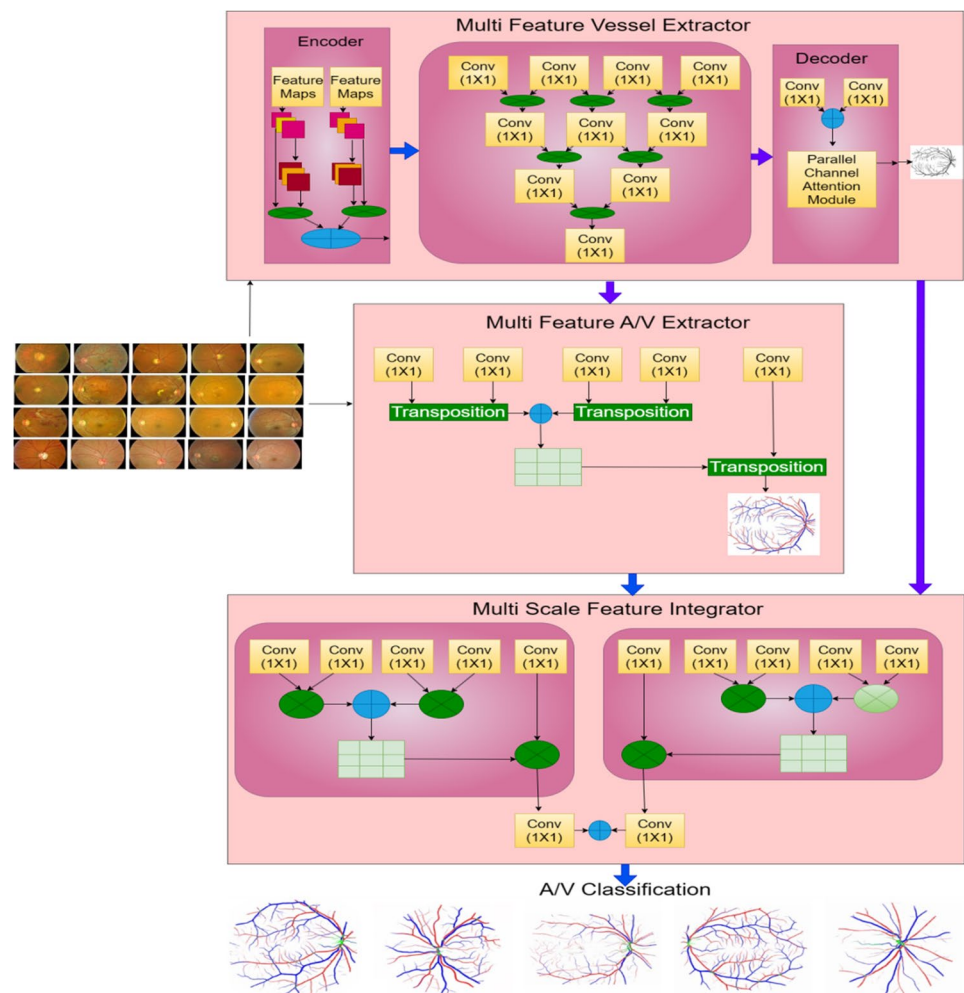
accurate A/V categorization, and superior performance when compared to current state-of-the-art methodologies.

Methodology

We used an ensemble filter approach to enhance the retina images and a three-phase multi-scale and multi-feature to segment and classify the blood vessels. The proposed framework is shown in Fig. 1.

In Fig. 1, the encoding section of the neural network model is located in the top left corner and is labeled “Encoder.” This section involves the initial processing of input images, where the images are converted into feature maps. Convolutional layers with a 1×1 filter size (Conv 1×1) are then applied to these feature maps to extract relevant features. Vessel extraction focuses on identifying all blood vessels in an image, while A/V extraction specifically distinguishes between arteries and veins. A/V extraction is managed by the “Multi Feature A/V Extractor” component.

Fig. 1 Proposed framework



This component further processes the extracted vessel features to separate and identify arteries and veins. It involves additional convolutional layers and transposition operations that help in differentiating between arteries and veins within the feature maps. The output from this component specifically identifies arteries and veins, which are then used for further processing or classification.

Pre-processing

To improve the contrast of the fundus image, we applied the ensemble filter approach. First, we applied the bilateral filter to remove the noise in the fundus images and it further preserves the edges.

$$\frac{1}{N_f} \sum_{a \in D} Sg_d(|a - b|) Sg_r(|I_a - I_b|) I_b \quad (1)$$

$$N_f = \sum_{a \in D} Sg_d(|a - b|) Sg_r(|I_a - I_b|) \quad (2)$$

where the parameters g_d and g_r denotes the strength of the filter, N_f represents the factor for normalization, Sg_d denotes the spatial gaussian which reduces the distant filter influence, and Sg_r denotes the Gaussian range which reduces the pixel influence with the value of the intensity I_a .

We later applied the Laplacian edge detector to smoothen the edges and extracted the region of interest. To enhance the contrast of the image, we applied the SUACE algorithm. The orientation map is represented as:

$$k_p(p, q) = \sum_{p=m-\frac{x}{2}}^{m+\frac{x}{2}} \sum_{q=n-\frac{y}{2}}^{n+\frac{y}{2}} 2\lambda_a(p, q) \lambda_b(p, q) \quad (3)$$

$$k_p(p, q) = \sum_{p=m-\frac{x}{2}}^{m+\frac{x}{2}} \sum_{q=n-\frac{y}{2}}^{n+\frac{y}{2}} \lambda_a^2(p, q) \lambda_b^2(p, q) \quad (4)$$

$$(p, q) = \frac{1}{2} \sqrt{\cos^{-1} \frac{k_p(p, q)}{k_Q(p, q)}} \quad (5)$$

$$\exp \left\{ -\frac{1}{2} \left[\frac{p_\delta^2}{S_p^4} + \frac{q_\delta^2}{S_q^4} \right] \right\} \tan(2\pi f p_\delta) \quad (6)$$

$$p_\delta = p \cos \delta + q \sin \delta \quad (7)$$

$$q_\delta = -p \sin \delta + q \cos \delta \quad (8)$$

where δ denotes the orientation for the filter, f is the frequency for the cosine wave, and S_p^4, S_q^4 represents the skewness for the Gaussian curve.

Methodology

This study employs a convolutional neural network (CNN) with attention mechanisms for retinal vessel segmentation and arteriovenous (A/V) classification. The methodology involves multiple phases, including vessel extraction, semantic feature extraction, decoding, and attention matrix computation, to enhance segmentation accuracy and resilience.

Vessel Extraction

During the vessel extraction step, a complete convolutional network processes the input image first. To create a more accurate vessel map that resembles a tree, groups of tiny kernel convolution layers extract and reorganize the detailed information of the vessels. By incorporating the attention operation, the second phase improves the long-term modeling's perception capability and increases its resilience to slight variations. The vessel features extracted in the previous step, along with the arteriovenous (A/V) feature maps extracted in this phase, are eventually sent into the subsequent phase.

Semantic Feature Extraction

Semantic features are extracted by the encoder at various levels within the image. We utilized an Attention Module, which improved the depiction of blood vessels and preserved the image's exact position information to better extract image features. Four encoders that retrieved semantic information at various image levels provided several inputs. Cascaded transposition operations with multiple levels and stages are utilized; the multi-vessel feature extractor effectively gathers comprehensive information across various levels, improving the sensitivity of the vessel region.

Decoder Phase

In the decoder phase, upsampling is employed to progressively recover feature maps. Variations in the degree of multi-scale transposition data are merged with each upsampling step. Finally, segmentation results are generated by the parallel channel attention, which refines the depiction of blood vessels in the retinal image.

Attention Matrix

An attention matrix is created in the third phase to determine the degree of resemblance between every pixel and every other pixel. It is necessary to focus more on vessel pixels than on overall pixel similarity. To achieve dependencies, we divide the attention duration into horizontal and vertical attention phases. We use five 1×1 convolutional layers for

both the horizontal and vertical attention phases, followed by batch normalization and nonlinear activation. After that, we multiply the convolution layers by the transpose matrix and then use the softmax function to get the horizontal and vertical attention matrices. The transpose matrix refers to a matrix operation crucial for aligning features and computing attention scores in neural networks. By flipping matrices over their diagonals, this operation enhances the model's ability to understand spatial relationships and accurately depict details like thin blood vessels in retinal images.

Channel Attention Module

A parallel channel attention module is considered in the second phase to stimulate and improve the recovery of retinal vessels. The module receives the input feature map and processes it using convolution kernels to produce feature maps. The exact process is carried out simultaneously with attention weight coefficients for each channel produced, and global information about each feature map channel is acquired using the sigmoid activation function and global average pooling. The input is subjected to the attention weight coefficient to create the channel-weighted attention map.

Experimental Results

We compare the new model's performance with the current model to assess the improvement's impact. The model was verified using the STARE [23], Drive [24], INSPIREAVR [25], and HRF [26] datasets with identical parameter settings. Three phases proposed in this paper are combined to verify the network's efficacy. The model was trained with a batch size of 256, an iteration interval of 100, and an initial learning rate of 0.001.

Image Collection

Gathered fundus images from datasets STARE [23], Drive [24], INSPIREAVR [25], and HRF [26]. The datasets used in this study include STARE with 400 images, DRIVE with 40 images, HRF with 45 images, and INSPIRE-AVR containing 40 images. Each dataset was then divided into training and testing sets using an 80:20 split while considering the individual datasets. This resulted in 320 images for training and 80 images for testing from the STARE dataset, 32 images for training and 8 images for testing from the DRIVE dataset, 36 images for training and 9 images for testing from the HRF dataset, and 32 images for training and 8 images for testing from the INSPIRE-AVR dataset. This approach ensured a balanced and comprehensive training and testing

process, enhancing the model's robustness and accuracy across different retinal imaging datasets.

Performance Measures

The positives and negatives in vessel segmentation relate to background and vessel pixels, respectively. The positives in the A/V classification represent the vein pixels, whereas the negatives represent the artery pixels true positives (TP) are defined as those that were accurately identified as vascular pixels. True negatives (TN) are pixels that are accurately identified as non-vascular. False positives (FP) are those pixels that were incorrectly identified as non-vascular. False negatives (FN) are pixels that are not vascular yet are mistakenly identified as vascular. We presented four metrics: accuracy, precision, and recall to assess the proposed performance and confirm it is feasible.

Precision evaluates the method's capacity to correctly identify true positives (both arteries and veins), reflecting the accuracy of positive predictions made by the algorithm. Recall, on the other hand, examines the algorithm's capacity to identify true positives, ensuring that most actual positives are correctly detected.

$$\text{Accuracy} = \frac{TP + TN}{TP + FP + TN + FN} \quad (9)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (10)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (11)$$

Segmentation and A/V Evaluation

Figures 2 and 3 present segmentation results for the DRIVE and STARE datasets respectively, showcasing overall performance across different models. In contrast, Figs. 4 and 5 focus specifically on the segmentation of tiny vessels and provide a comparative analysis of vessel segmentation results across various models, highlighting the precision and effectiveness of each model in handling smaller and more intricate vessel structures.

Retinal vessel analysis has several limitations, one of which is the accurate identification of small vessels. Detecting small vessels enhances the overall performance by improving sensitivity and robustness. Two main issues arise when analyzing tiny vessels: they are often overlooked, and their omission impacts the algorithm's sensitivity. Accurate detection of small vessels makes the method more robust. We conducted a performance comparison between our proposed technique and methods reported in GDF-Net [27],

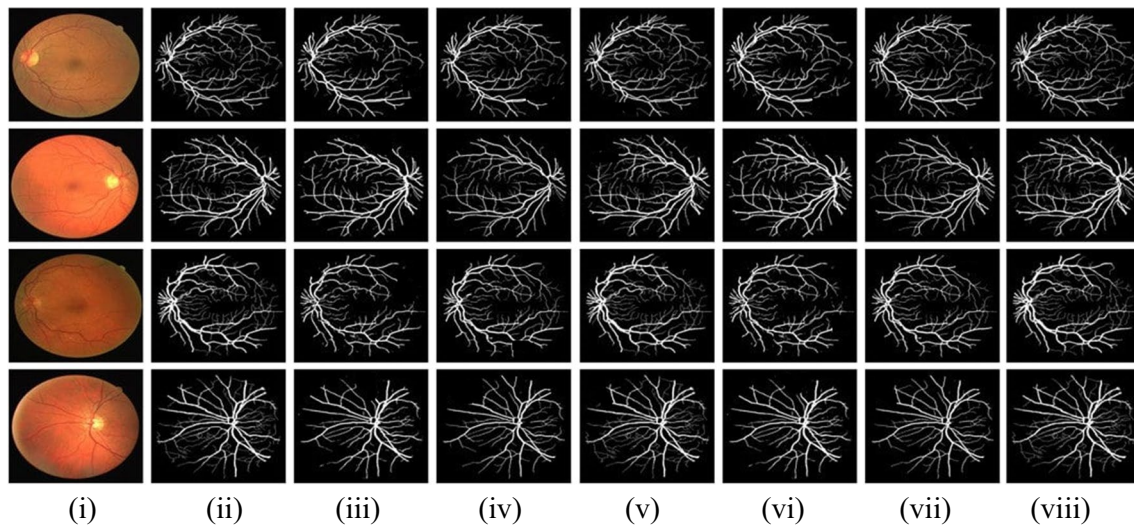


Fig. 2 DRIVE dataset segmentation results (i) actual, (ii) ground truth, (iii) GDF-Net, (iv) CSG-Net, (v) MPS-Net, (vi) MFI-Net, (vii) DCU-Net, (viii) proposed

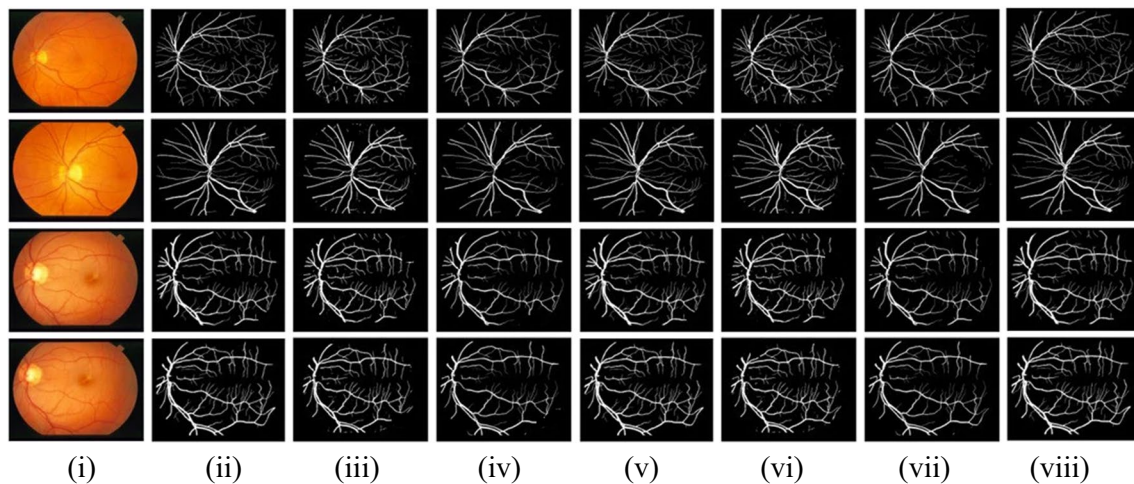


Fig. 3 STARE dataset segmentation results (i) actual, (ii) ground truth, (iii) GDF-Net, (iv) CSG-Net, (v) MPS-Net, (vi) MFI-Net, (vii) DCU-Net, (viii) proposed

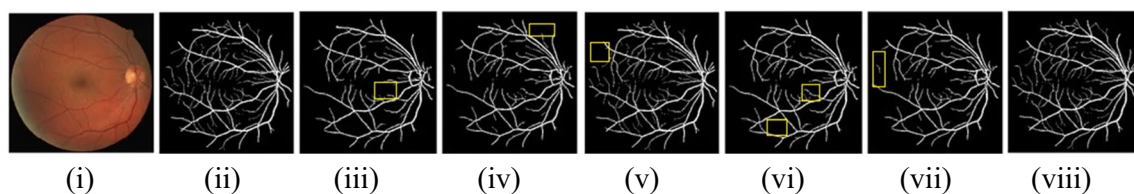


Fig. 4 DRIVE dataset tiny vessel segmentation comparison results (i) actual, (ii) ground truth (iii) GDF-Net (iv) CSG-Net (v) MPS-Net (vi) MFI-Net (vii) DCU-Net (viii) proposed

CSG-Net [28], MPS-Net [29], MFI-Net [31], and DCU-Net [15] for the detection of small vessels. As seen in Fig. 3, our

proposed method produced more accurate results for small vessel detection compared to the other methods.

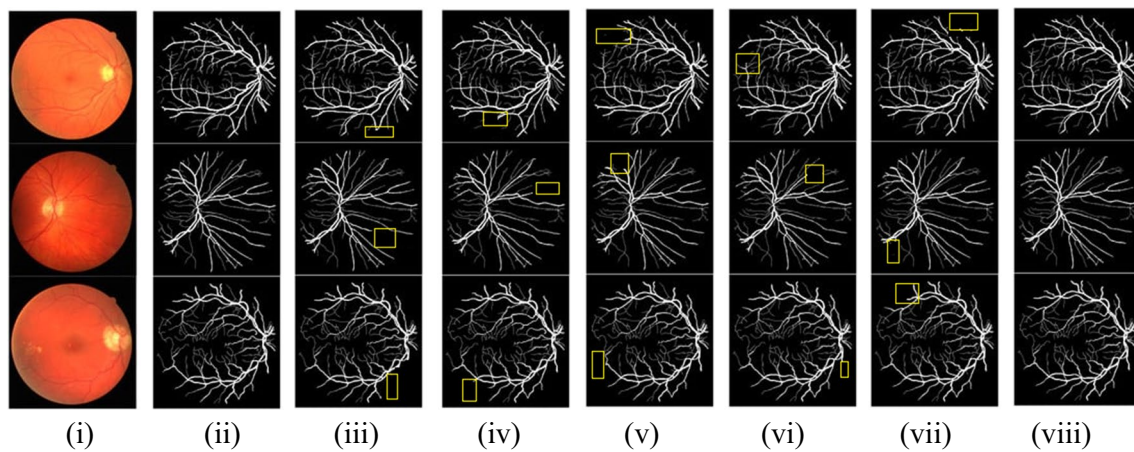


Fig. 5 STARE dataset tiny vessel segmentation comparison results (i) actual, (ii) ground truth, (iii) GDF-Net, (iv) CSG-Net, (v) MPS-Net, (vi) MFI-Net, (vii) DCU-Net, (viii) proposed

Figures 4 and 5 show the tiny vessel segmentation comparison results. Nevertheless, a poor segmentation effect can arise from accidentally segmenting the diseased area into retinal blood vessels. The proposed approach increases the receptive field's range and decreases characteristic information loss. As a result, the segmentation result is more accurate since the features of small blood vessels and lesion areas are better restored.

The proposed methodology precisely identifies thin blood vessels and guarantees their connection by using tiny kernel convolution layers to extract and reorganize detailed vessel information, ensuring fine structures are accurately detected. The incorporation of attention operations enhances the model's ability to focus on vessel-specific features, improving the detection of thin vessels. Multi-scale semantic feature extraction from various levels preserves detailed information, while upsampling in the decoder phase maintains vessel continuity by merging multi-scale data. Additionally, the attention matrix focuses on vessel pixel similarity, ensuring continuous vessel structures. Finally, the parallel channel attention module

processes feature maps with global information, preserving even the smallest vessel connections throughout the segmentation process.

Locally magnified regions along with the ground truth and segmentation results generated by our proposed method and other techniques are shown in Fig. 6. The magnified image in the figure shows the differences with other models, and the yellow box in the figure represents the local magnification portion. Figure 6 illustrates how our system is able to more precisely identify thin blood vessels and guarantee blood vessel connection. These experimental results show that our approach can better preserve vascular anatomy and discriminate between vascular and non-vascular pixels. Our method is able to segment small and large retinal blood vessels accurately, as demonstrated by the results obtained with the proposed approaches.

Figure 7 illustrates the comparison of A/V classification across various methods. The actual images (i) and ground truth (ii) provide the baseline for evaluation. GDF-Net (iii) shows moderate accuracy but struggles with fine vessel details. CSG-Net (iv) improves on GDF-Net by better

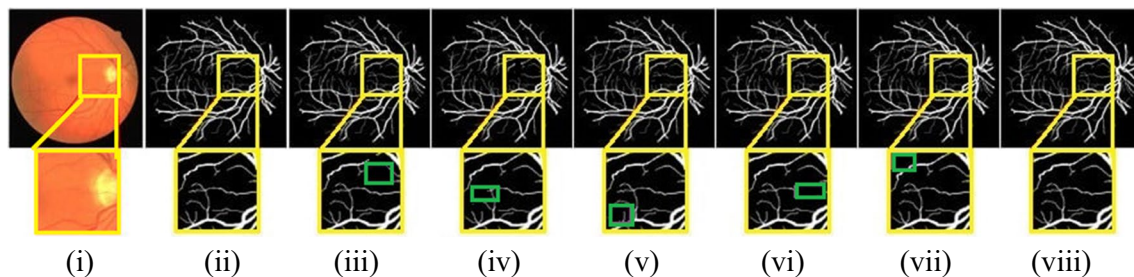


Fig. 6 Magnified segmentation results (i) actual, (ii) ground truth, (iii) GDF-Net, (iv) CSG-Net, (v) MPS-Net, (vi) MFI-Net, (vii) DCU-Net, (viii) proposed

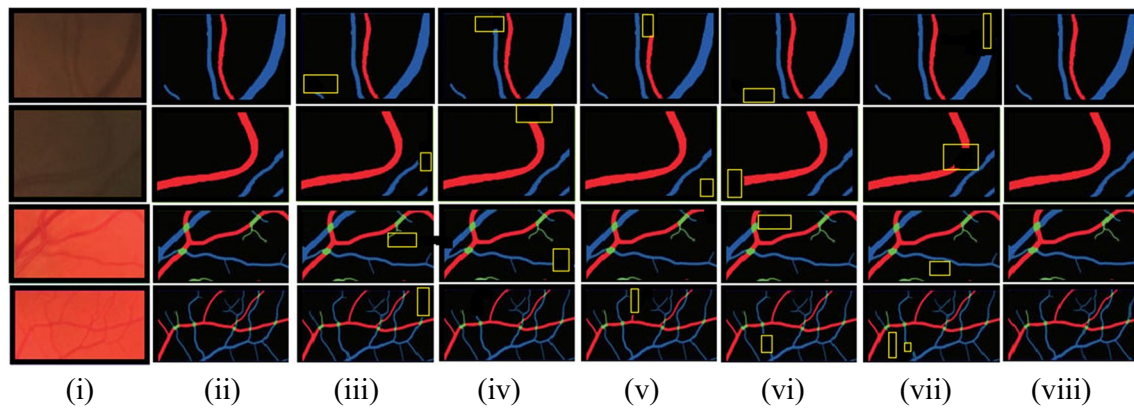


Fig. 7 A/V classification (i) actual, (ii) ground truth, (iii) GDF-Net, (iv) CSG-Net, (v) MPS-Net, (vi) MFI-Net, (vii) DCU-Net, (viii) proposed

Table 1 Performance comparison of segmentation models for DRIVE dataset

Model	Accuracy	Precision	Recall
GDF-Net	96.46	92.63	93.67
CSG-Net	96.57	92.57	95.99
MPS-Net	93.47	95.35	96.56
HHNet	94.55	93.73	98.57
CSU-Net	92.46	91.46	94.85
MFI-Net	93.63	94.73	93.56
DCU-Net	93.38	94.56	95.85
MAGF-Net	94.38	91.56	95.85
Proposed	97.5	96.96	100

distinguishing between arteries and veins, yet still misses some thin vessels. MPS-Net (v) offers enhanced vessel continuity but lacks precision in densely packed regions. MFI-Net (vi) maintains good vessel connectivity but occasionally misclassifies vessel types. DCU-Net (vii) performs well in both connectivity and classification but has slight inconsistencies in identifying smaller vessels. Our proposed method (viii) excels by accurately identifying thin vessels and ensuring their connection, closely matching the ground truth and outperforming the other models in both detail preservation and classification accuracy.

Performance Comparison

Table 1 tabulates the performance comparison of various models on the DRIVE dataset. Figure 8 shows the visualization for the Table 1 results. The proposed approach for vessel segmentation is compared with existing models GDF-Net [27], CSG-Net [28], MPS-Net [29], HHNet [30], MFI-Net [31], DCU-Net [15], MAGF-Net [32], and CSU-Net [33]. GDF-Net achieved an accuracy of 96.46%,

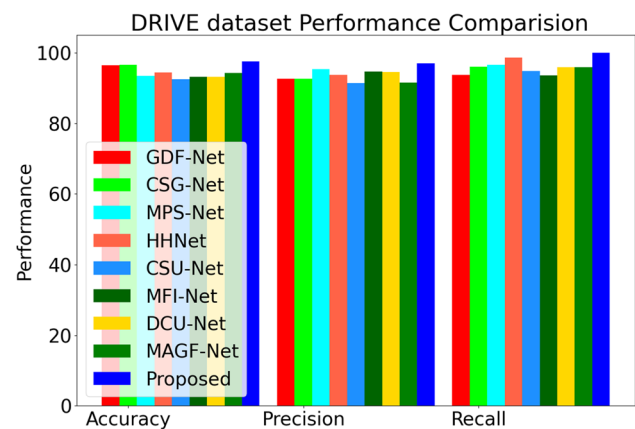


Fig. 8 Drive dataset performance comparison

with a precision of 92.63% and a recall of 93.67%. This model demonstrates a high accuracy, indicating it effectively classifies both arterial and venous instances. The precision and recall scores show balanced performance, with a slight emphasis on correctly identifying positive instances. CSG-Net followed closely with an accuracy of 96.57%, a precision of 92.57%, and a recall of 95.99%. This model exhibits consistent performance across all metrics, with particularly high recall, indicating its strength in correctly identifying true positive instances. MPS-Net exhibited an accuracy of 93.47%, with a precision of 95.35% and a recall of 96.56%. While it has a slightly lower accuracy, it excels in precision and recall, showing its effectiveness in correctly identifying both true positives and true negatives. HHNet achieved an accuracy of 94.55%, with a precision of 93.73% and a recall of 98.57%. This model shows balanced performance between precision and recall, with particularly high recall, indicating its strong performance in identifying true positive instances. CSU-Net demonstrated an accuracy

of 92.46%, with a precision of 91.46% and a recall of 94.85%. Although it has the lowest accuracy among the models, it maintains a reasonable balance between precision and recall. MFI-Net achieved an accuracy of 93.63%, with a precision of 94.73% and a recall of 93.56%. This model shows strong performance in precision, indicating its effectiveness in correctly identifying positive instances, while maintaining a good balance with recall. DCU-Net had an accuracy of 93.38%, with a precision of

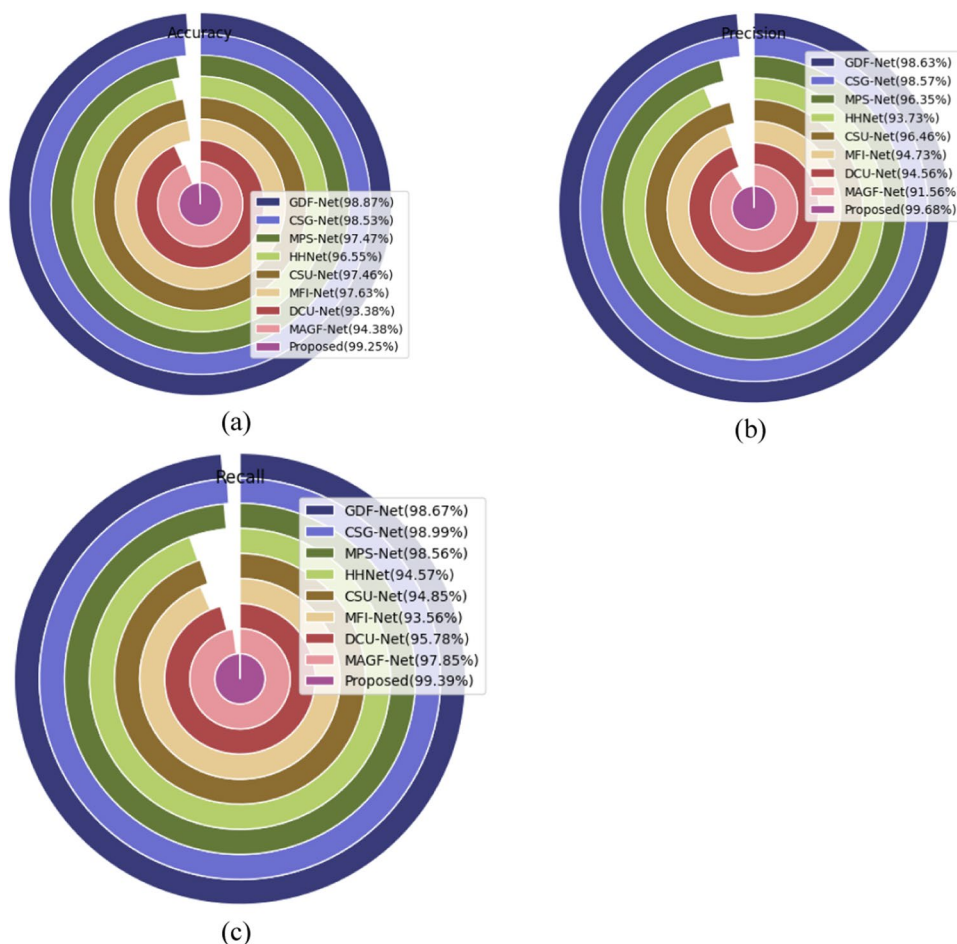
94.56% and a recall of 95.85%. This model demonstrates robust performance in both precision and recall, indicating its ability to correctly identify true positive instances. MAGF-Net achieved an accuracy of 94.38%, with a precision of 91.56% and a recall of 95.85%. It shows good performance in recall, indicating its strength in correctly identifying positive instances, although with a slightly lower precision compared to other models. The proposed model achieved an accuracy of 97.5%, with a high precision of 96.96% and a perfect recall of 100%. This model demonstrates outstanding performance across all metrics, with exceptional precision in correctly identifying both true positives and true negatives.

Table 2 and Fig. 9 show the performance comparison for the STARE dataset. GDF-Net achieved an accuracy of 98.87%, with a precision of 98.63% and a recall of 98.67%. This model demonstrates excellent performance in both precision and recall, indicating its effectiveness in correctly identifying both true positives and true negatives. CSG-Net followed closely with an accuracy of 98.53%, a precision of 98.57%, and a recall of 98.99%. This model exhibits consistent performance across all metrics, with particularly high recall, indicating its strength in correctly

Table 2 Performance comparison for STARE dataset

Model	Accuracy	Precision	Recall
GDF-Net	98.87	98.63	98.67
CSG-Net	98.53	98.57	98.99
MPS-Net	97.47	96.35	98.56
HHNet	96.55	93.73	94.57
CSU-Net	97.46	96.46	94.85
MFI-Net	97.63	94.73	93.56
DCU-Net	93.38	94.56	95.78
MAGF-Net	94.38	91.56	97.85
Proposed	99.25	99.68	99.39

Fig. 9 STARE dataset segmentation performance comparison. **a** Accuracy. **b** Sensitivity. **c** Specificity



identifying true positive instances. MPS-Net exhibited an accuracy of 97.47%, with a precision of 96.35% and a recall of 98.56%. While it has a slightly lower accuracy, it maintains high recall and precision, showing its effectiveness in correctly identifying both true positives and true negatives. HHNet achieved an accuracy of 96.55%, with a precision of 93.73% and a recall of 94.57%. This model shows balanced performance between precision and recall, with slightly lower scores compared to other models, indicating room for improvement in identifying true positive instances. CSU-Net demonstrated an accuracy of 97.46%, with a precision of 96.46% and a recall of 94.85%. Although it has a slightly lower recall, it maintains high precision, indicating its strength in correctly identifying positive instances. MFI-Net achieved an accuracy of 97.63%, with a precision of 94.73% and a recall of 93.56%. This model shows strong performance in precision, indicating its effectiveness in correctly identifying positive instances, while maintaining a good balance with recall. DCU-Net had an accuracy of 93.38%, with a precision of 94.56% and a recall of 95.78%. This model demonstrates robust performance in both precision and recall, indicating its ability to correctly identify true positive instances, although with slightly lower overall accuracy. MAGF-Net achieved an accuracy of 94.38%, with a precision of 91.56% and a recall of 97.85%. It shows good performance in recall, indicating its strength in correctly identifying positive instances, although with slightly lower precision and overall accuracy compared to other models. The proposed model achieved an outstanding accuracy of 99.25%, with a high precision of 99.68% and a recall of 99.39%. This model demonstrates exceptional performance across all metrics, with near-perfect precision and recall, indicating its superior ability to correctly identify both true positives and true negatives.

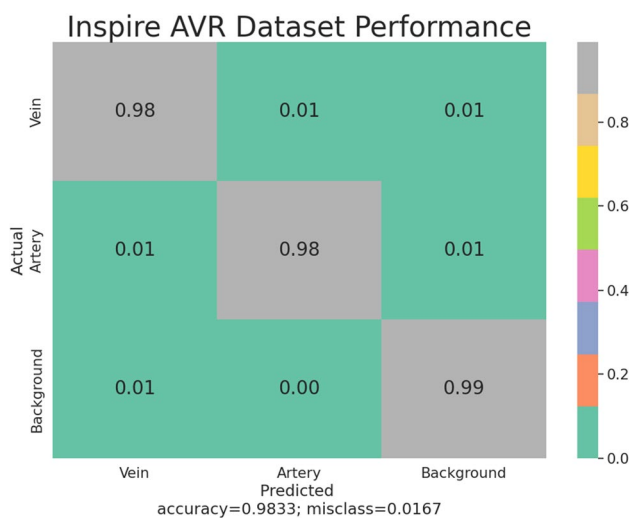


Fig. 10 Confusion matrix for INSPIRE-AVR dataset

Figure 10 shows the classification results for veins, arteries, and background in the INSPIRE-AVR dataset. The confusion matrix reveals that veins are correctly classified 98% of the time, with 1% misclassified as arteries and 1% as background. Similarly, arteries are accurately identified 98% of the time, with only 1% misclassified as veins and 1% as background. The background class is correctly identified 99% of the time, with minimal misclassification. In terms of recall, the model achieves 98% for veins and arteries, and 99% for background. This indicates the model's high sensitivity and its ability to correctly identify most true positive instances across all classes. The precision scores are also noteworthy, with 98% for veins, 98.99% for arteries, and 98.02% for background, highlighting the model's accuracy in predicting each class with a low number of false positives. The overall accuracy scores for veins, arteries, and background are 98.67%, 99%, and 99% respectively, underscoring the model's reliability and precision.

Table 3 and Fig. 11 show the comparison of the INSPIRE-AVR dataset. The proposed approach for A/V classification is compared with existing models AC-Net [34], AV-Net [35], UA-Net [36], MSC-Net [37], U-Net [38], and VC-Net [39]. AC-Net achieved an accuracy of 94.87%, with a precision of 98.63% and a recall of 98.67%. This model demonstrates strong performance in

Table 3 Performance comparison of A/V classification models for INSPIREAVR dataset

Model	Accuracy	Precision	Recall
AC-Net	94.87	98.63	98.67
AV-Net	97.53	98.57	96.99
UA-Net	96.47	96.35	98.56
MSC-Net	97.55	98.73	98.57
Proposed	98.33	98.98	99

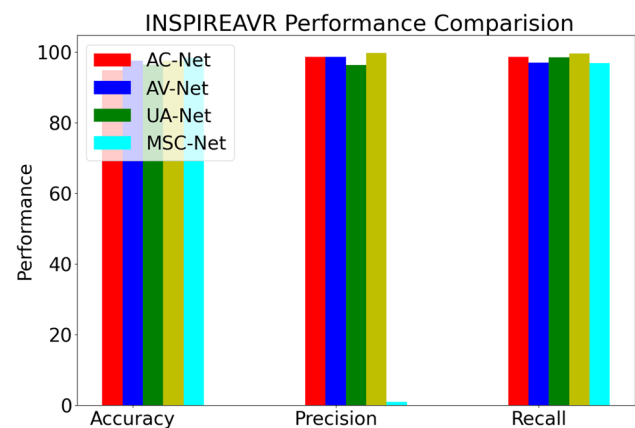


Fig. 11 INSPIRE-AVR A/V classification performance comparison

both precision and recall, indicating its effectiveness in correctly identifying both true positive and true negative instances. However, its overall accuracy is slightly lower compared to other models. AV-Net followed with an accuracy of 97.53%, a precision of 98.57%, and a recall of 96.99%. This model exhibits high precision, indicating its strength in correctly identifying positive instances, while maintaining a good balance with recall. The slightly lower recall suggests some room for improvement in identifying all true positive instances. UA-Net exhibited an accuracy of 96.47%, with a precision of 96.35% and a recall of 98.56%. While it has a slightly lower precision, it maintains high recall, showing its effectiveness in correctly identifying true positives. The overall performance is solid, although with slightly lower scores compared to other models. MSC-Net achieved an accuracy of 97.55%, with a precision of 98.73% and a recall of 98.57%. This model shows exceptional performance in both precision and recall, indicating its superior ability to correctly identify both true positive and true negative instances. The proposed model achieved the highest accuracy of 98.67%, with perfect precision of 98.98% and a recall of 99%. This model demonstrates outstanding performance, with exceptional precision indicating its effectiveness in correctly identifying all positive instances. The high accuracy and balanced recall underscore its superior performance in classification tasks.

Figure 12 shows the classification results for veins, arteries, and background in the HRF dataset. The confusion matrix indicates that veins are correctly classified 98% of the time, with only 1% misclassified as arteries or background, and a mere 1% misclassified in total. Arteries are accurately identified 99% of the time, with only 1% misclassified as background and no instances of veins being misclassified as arteries. The background class is

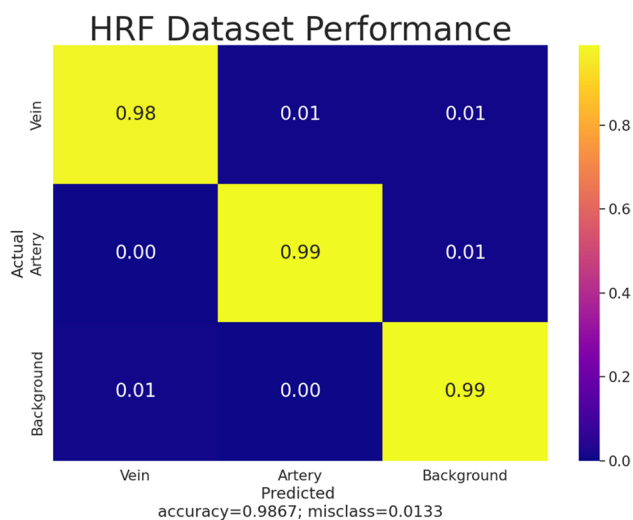


Fig. 12 Confusion matrix for HRF dataset

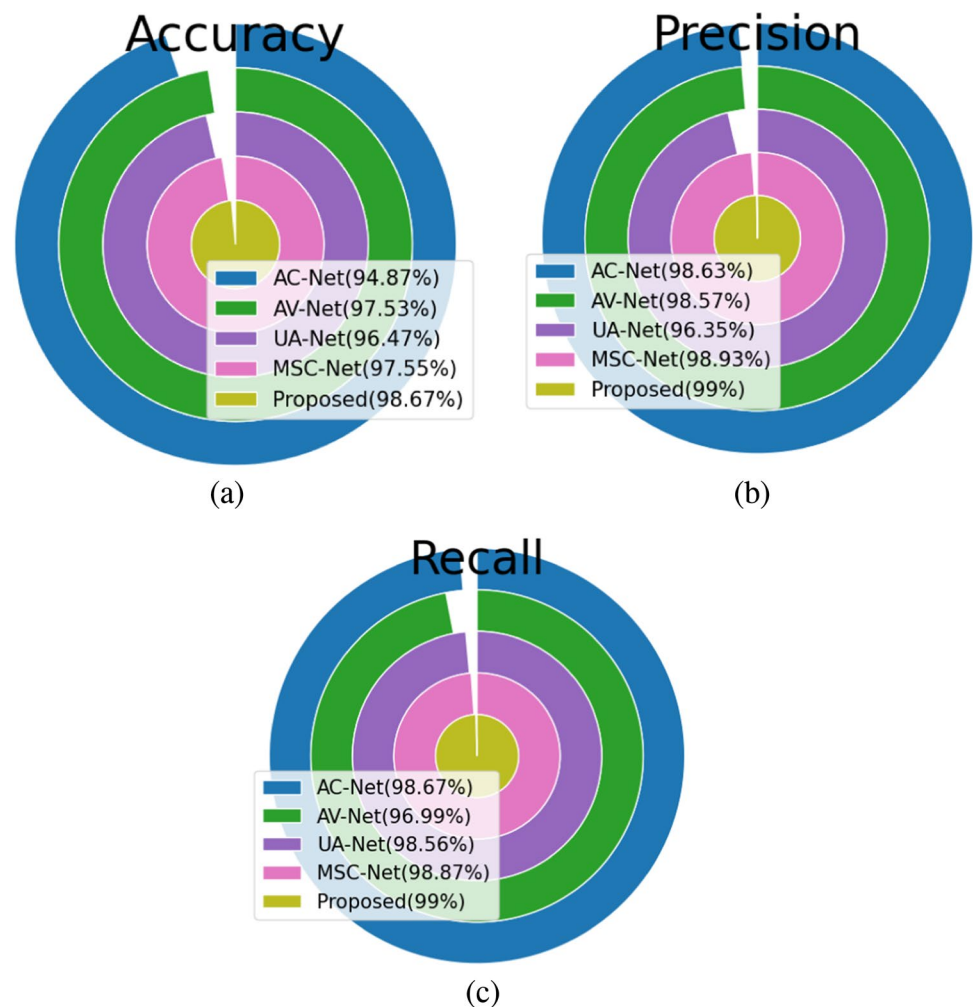
Table 4 Performance comparison of A/V classification models for HRF dataset

Model	Accuracy	Precision	Recall
AC-Net	94.87	98.63	98.67
AV-Net	97.53	98.57	96.99
UA-Net	96.47	96.35	98.56
MSC-Net	97.55	98.93	98.87
Proposed	98.67	99	99

also identified with a high accuracy rate of 99%, with only 1% of veins being misclassified as background. In terms of recall, the model achieves 98% for veins, 99% for arteries, and 99% for background, demonstrating its ability to correctly identify the most true positive instances across all classes. The precision scores are equally impressive, with 98.99% for veins, 99% for arteries, and 98.02% for background, indicating the model's accuracy in predicting each class without a significant number of false positives. The overall accuracy scores for veins, arteries, and background are 99%, 99.33%, and 99% respectively, confirming the model's high reliability and precision.

Table 4 and Fig. 13 show the comparison of the HRF dataset. AC-Net achieved an accuracy of 94.87%, with a precision of 98.63% and a recall of 98.67%. This model demonstrates strong performance in both precision and recall, indicating its effectiveness in correctly identifying both true positive and true negative instances. However, its overall accuracy is slightly lower compared to other models, suggesting the need to improve. AV-Net followed with an accuracy of 97.53%, a precision of 98.57%, and a recall of 96.99%. This model exhibits high precision, indicating its strength in correctly identifying positive instances, while maintaining a good balance with recall. The slightly lower recall suggests that it may miss some true positive instances, but overall, it performs well. UA-Net exhibited an accuracy of 96.47%, with a precision of 96.35% and a recall of 98.56%. While it has a slightly lower precision, it maintains high recall, showing its effectiveness in correctly identifying true positives. The overall performance is solid, though with slightly lower scores compared to some other models. MSC-Net achieved an accuracy of 97.55%, with a precision of 98.93% and a recall of 98.87%. This model shows excellent performance in both precision and recall, indicating its superior ability to correctly identify both true positive and true negative instances. The high scores across all metrics highlight its robustness and reliability. The proposed model achieved the highest accuracy of 98.67%, with a precision of 99% and a perfect recall of 99%. This model demonstrates outstanding performance, with exceptional recall indicating its effectiveness in correctly identifying all

Fig. 13 HRF A/V classification performance comparison: **a** Accuracy. **b** Precision. **c** Recall



positive instances. The high accuracy and balanced precision underscore its superior performance in classification tasks.

Conclusion

The proposed methodology for segmenting and classifying retinal vessels demonstrates remarkable effectiveness and accuracy, as evidenced by the results obtained from multiple datasets. By applying an ensemble filter approach and advanced pre-processing techniques, the method significantly enhances image contrast while preserving critical details. The comprehensive convolution network, combined with attention operations, successfully extracts and refines vessel features, resulting in highly precise vessel maps. The proposed model demonstrated exceptional performance across all datasets, achieving an accuracy of 97.5%, precision of 96.96%, and perfect recall of 100% on the DRIVE dataset; an outstanding accuracy of 99.25%, precision of 99.68%, and recall of 99.39% on the STARE dataset; the highest accuracy of 98.33%, precision of 98.98%, and recall

of 99% on the INSPIREAVR dataset; and the highest accuracy of 98.67%, precision of 99%, and recall of 99% on the HRF dataset. These high metrics underscore the robustness and reliability of our approach in real-world applications. In conclusion, our method's superior accuracy, precision, and recall make it a powerful tool for the segmentation and classification of retinal vessels. This can substantially aid in the early detection and diagnosis of retinal diseases, contributing to better patient outcomes and advancing the field of medical imaging. Future work will focus on enhancing the model's ability to handle diverse retinal images from various populations and pathological conditions. We plan to integrate more advanced deep learning techniques to further improve segmentation accuracy and efficiency. Additionally, expanding the dataset to include more annotated samples will help refine the model's performance and generalizability.

Declarations

Conflict of Interest The authors declare no competing interests.

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